

Business Question and Visualization Report

Phase: Data Visualization

Project Name: Food Consumer Behaviour Analysis

Team ID: PNT2022TMID01234

Date: 24 September 2024 / 15 October 2024

Maximum Marks: 5 Marks

Visualization Development Overview

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting.

Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations in Tableau. Well-framed questions help in identifying key metrics, selecting relevant data, and building visualisation that provide insights.

1. What are the demographic spending trends across age cohorts and income tiers?

Visualization: Clustered Column & Dual-Axis Line Chart comparing Average Meal Spend (\$) and Order Frequency across Age Cohorts (18-24, 25-34, 35-44, 45-54, 55+).

Key Insight / Observation: The 25-34 age demographic represents the highest cumulative expenditure and ordering volume, while the 45-54 high-income cohort shows the highest individual ticket size (\$62.40/order).

Visualization Screenshot: *Embedded in live Tableau Public Workbook*

2. How is customer dining demand distributed across ordering channels?

Visualization: Donut Chart & 100% Stacked Bar displaying the market share breakdown of Delivery Apps (54.2%), Dine-in (29.6%), and Takeout (16.2%).

Key Insight / Observation: Digital delivery apps dominate weekly order volume, particularly among urban millennials, whereas dine-in retains higher average spend per sitting on weekends.

Visualization Screenshot: *Embedded in live Tableau Public Workbook*

3. Which cuisine categories generate the highest total revenue and order volume?

Visualization: Horizontal Ranked Bar Chart sorting total revenue contribution by cuisine family (Fast Food, Asian, Italian, Healthy/Organic, Mexican, Desserts).

Key Insight / Observation: Fast Food and Asian cuisines lead total volume, but Healthy/Organic cuisines show the fastest year-over-year revenue growth (+28%) and premium average ticket sizes.

Visualization Screenshot: *Embedded in live Tableau Public Workbook*

4. What are the seasonal and monthly dining expenditure trends?

Visualization: Time-Series Multi-Line Chart tracking monthly sales and transaction volume from January through December.

Key Insight / Observation: Pronounced spending peaks occur in November and December (holiday festivities and promotional campaigns), with a secondary surge in May, while February records the lowest seasonal volume.

Visualization Screenshot: *Embedded in live Tableau Public Workbook*

5. How does delivery duration impact customer satisfaction ratings?

Visualization: Scatter Plot with Trend Line & Heatmap plotting Delivery Time (minutes) against Customer Rating (1.0 to 5.0 stars).

Key Insight / Observation: Deliveries completed within 30 minutes maintain a 4.7/5.0 average rating; satisfaction drops precipitously to 2.8/5.0 when transit times surpass 45 minutes.

Visualization Screenshot: *Embedded in live Tableau Public Workbook*

6. What is the effect of promotional discounts on Average Order Value (AOV)?

Visualization: Bubble Chart plotting Discount Depth (0% to 40%) vs. AOV vs. Total Customer Count.

Key Insight / Observation: Moderate discounts (15-20%) maximize overall revenue by stimulating larger basket sizes, whereas discounts above 30% erode gross margins without proportional volume gains.

Visualization Screenshot: *Embedded in live Tableau Public Workbook*

7. How does household size and family composition influence meal type and channel choice?

Visualization: Segmented Treemap showing order frequency and preferred meal type (Solo Lunch, Family Dinner, Weekend Brunch) by Household Size (1, 2, 3-4, 5+).

Key Insight / Observation: Single-member households order 4.8x/week primarily via delivery apps, while households of 3+ members favor weekend dine-in and multi-item family bundles.

Visualization Screenshot: *Embedded in live Tableau Public Workbook*

8. How does customer satisfaction and dining spend vary across geographic regions/city tiers?

Visualization: Filled Geographic Map visualization showing regional customer density, average ticket spend, and Net Promoter Score (NPS) across urban tiers.

Key Insight / Observation: Tier 1 metropolitan zones demonstrate 68% delivery penetration and higher discretionary spend (\$48.50 avg), while Tier 2 regions show strong dine-in loyalty and higher average satisfaction (4.6/5.0).

Visualization Screenshot: *Embedded in live Tableau Public Workbook*

Note: Min 8 business question and visualisations Required inform of above